Day 9

Accuracy of Systems

The ability of the system follow the given instruction can be used as a measure of classification

* Common input : step , ramp, parabolic

$u(t)=1$, $r(t)=t$, $P(t)=t^2$



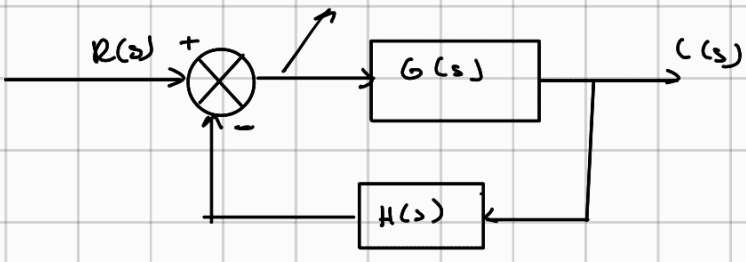


* Comparison of accuracy is possible by using the steady-state error magnitude !!

if $\zeta < 1$ in a second o.s



$E(s) = R(s) - C(s)H(s)$



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$E(s) = R(s) - G(s)H(s) \Rightarrow \frac{E(s)}{R(s)} = 1 - \frac{G(s)H(s)}{R(s)}$$

$$\Rightarrow \frac{E(s)}{R(s)} = 1 - \frac{G(s)H(s)}{1+G(s)H(s)} = \frac{1}{1+G(s)H(s)}$$

Final value theorem

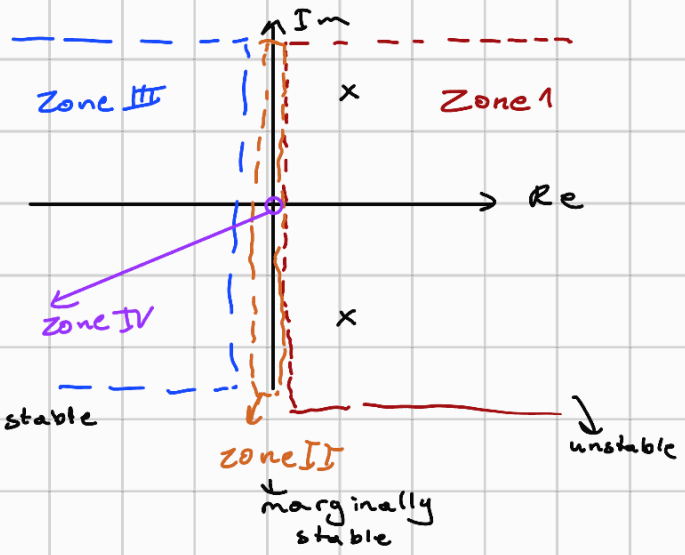
$$e(\infty) = e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)H(s)}$$

Validity of final value theorem:

If the poles are in zone 1

so it can't have final value

Only works with Zone III and zone IV



Stability → CLP ✓

OLP

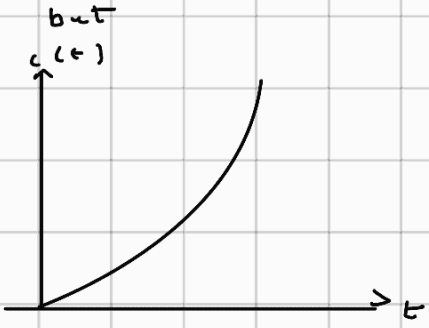
Zone I :

$$G(s) = T.F = \frac{1}{s-2}$$

$$FVT = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s}{s-2} = 0$$

✗ wrong

but

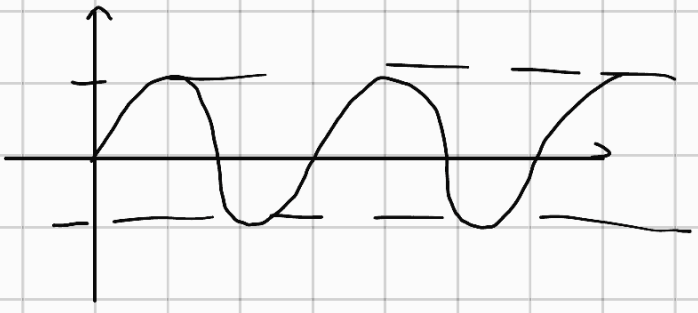


For Zone II

$$T.F = G(s) = \frac{1}{s^2+2}$$

$$s_{1,2} = \pm j\sqrt{2}$$

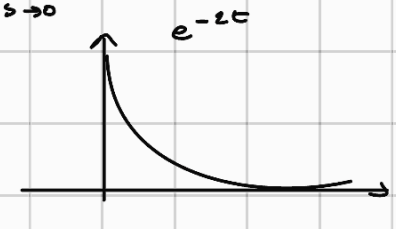
$$FVT: \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s}{s^2+2} = 0$$ ✗



Zone III

$$T.F = G(s) = \frac{1}{s+2} \quad \Rightarrow s = -2$$

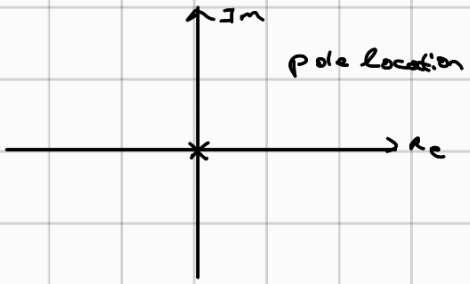
$$FVT: \lim_{s\to 0} \frac{s}{s+2} = 0$$



Zone IV:

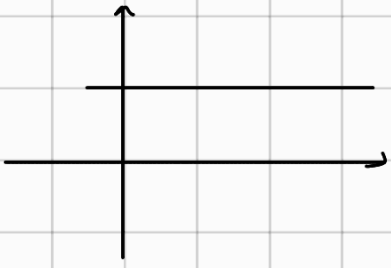
$TF = \frac{1}{s}$

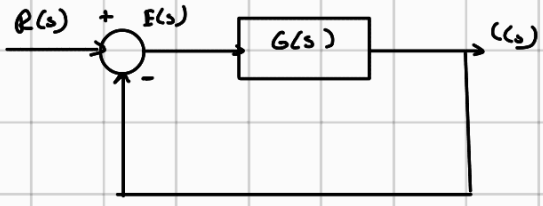
$FVT = \lim_{s \to 0} s \frac{1}{s} = 1$



$\lambda = 0$

$e^{\lambda t} = e^{0t} = 1$

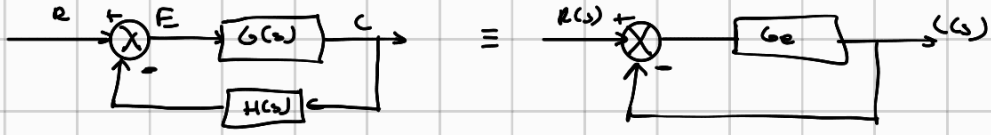




$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$

G(s): op on loop TF
for unity feedback

$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)}$



$$G(s) = \frac{k(T_a s+1)(T_b s+1)\cdots(T_m s+1)}{s^N (T_1 s+1)(T_2 s+1)\cdots(T_p s+1)}$$

$P_{1,\ldots,N} = 0$ ← N multiplicity → integrators

$P = -\frac{1}{T_1}, -\frac{1}{T_2}, \ldots$

if $N = 0$ → type zero

if $N = 1$ → Type one

⋮

In most cases the most integrator is 3

Accuracy of systems

case 1) step input: $R(s) = 1/s$

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{1}{1+G(s)} = \frac{1}{1+\lim_{s\to 0} G(s)} = \frac{1}{1+k_p}$$

constant and

$k_p = \lim_{s\to 0} G(s)$ (static position error) — DC gain

Type 0: $$G(s) = \frac{k(T_a s+1)(T_b s+1)\cdots(T_m s+1)}{s^N (T_1 s+1)(T_2 s+1)\cdots(T_p s+1)}$$

$N=0 \Rightarrow k_p = \lim_{s\to 0} G(s) = k$

$N=1 \Rightarrow k_p = \lim_{s\to 0} G(s) = \infty$

$N=2 \Rightarrow k_p = \lim_{s\to 0} = \infty$

$N \geq 1 \Rightarrow k_p = \infty$

case 2) input is ramp

$R(s) = 1/s^2$

$$e_{ss} = \lim_{s\to 0} \frac{s\frac{1}{s^2}}{1+G(s)} = \lim_{s\to 0} \frac{1}{s(1+G(s))}$$

$$= \frac{1}{\lim_{s\to 0} sG(s)} = \frac{1}{k_v}$$

k_v = static velocity error $= \lim_{s\to 0} sG(s)$

Type 0: $k_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{sk(T_a s+1)(T_b s+1)\cdots(T_m s+1)}{(T_1 s+1)(T_2 s+1)\cdots(T_p s+1)} = 0$

Type 1: $k_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{k(T_a s+1)\cdots}{(T_1 s+1)} = k$

Type 2: $k_v = \lim_{s\to 0} \frac{s\,k(T_a s+1)\cdots}{s^2(T_1 s+1)\cdots} = \infty$

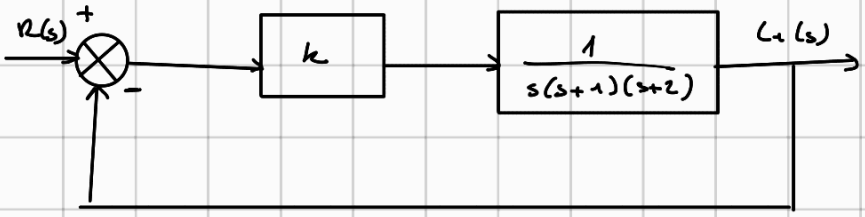
case 3)

$R(s) = 1/s^3$ $\quad e_{ss} = \lim_{s\to 0} \frac{s\frac{1}{s^3}}{1+G(s)} = \frac{1}{\lim_{s\to 0} s^2 G(s)} = \frac{1}{k_a}$ → static acceleration error

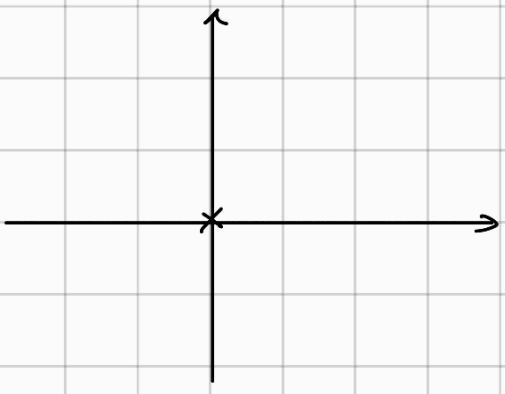
$N=0$: $k_a = \lim_{s\to 0} s^2 \frac{k(T_a s+1)(T_b s+1)\cdots}{(T_1 s+1)\cdots} = 0$

Type	input: step	ramp	parabolic
0	$\frac{1}{1+k}$	∞	∞
1	0	$1/k$	∞
2	0	0	$1/k$
3	0	0	0

Example:



$$OLT.F = \frac{k}{s(s+1)(s+2)}$$ Type 1 system



$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{s\cdot\frac{1}{s^2}}{1+\frac{k}{s(s+1)(s+2)}} = \lim_{s\to 0} \frac{s(s+1)(s+2)}{s[s(s+1)(s+2)+k]}$$

$$= \frac{2}{k} = 0{,}01 \Rightarrow k = 200 \checkmark$$

↑ for $e_{ss} = 0{,}01$ design

But we have to check stability

$$\frac{\frac{k}{s(s+1)(s+2)}}{1+\frac{k}{s(s+1)(s+2)}} = \frac{k}{s(s+1)(s+2)+k}$$

Stability $s(s+1)(s+2)+k=0$

$s(s^2+3s+2)+k=0$

kai p $s^3+3s^2+2s+k=0$

s^3	1	2
s^2	3	k
s^1	$\frac{6-k}{3}$	0
s^0	k	

$\rightarrow 6-k \geq 0 \Rightarrow k \leq 6$

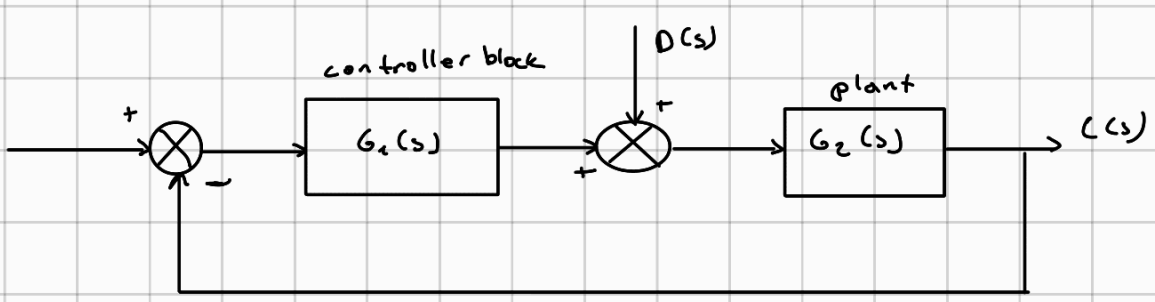
$k \geq 0$

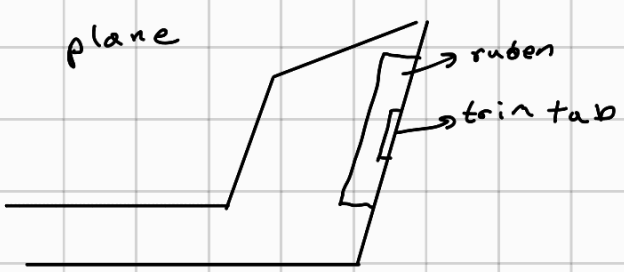
$0 \leq k \leq 6$

$k=6$ is the max possible k here !!!

so max $e_{ss}=0,33$ because the system have to be stable because final value theorem is not working with unstable systems.

Steady state error for disturbance





$$C(s) = [E(s)\,G_1(s) + D(s)]\,G_2(s)$$

$E(s) = R(s) - C(s) \Rightarrow C(s) = R(s) - E(s)$

$$E(s) = \frac{R(s)}{1+G_1(s)G_2(s)} - \frac{G_2(s)D(s)}{1+G_1(s)G_2(s)}$$

$$e_{ss} = \lim_{s\to 0} sE(s)$$

$$= \underbrace{\lim_{s\to 0} \frac{sR(s)}{1+G_1G_2}}_{e_R(\infty)} - \underbrace{\lim_{s\to 0} \frac{s\,G_2D}{1+G_1G_2}}_{e_D(\infty)}$$

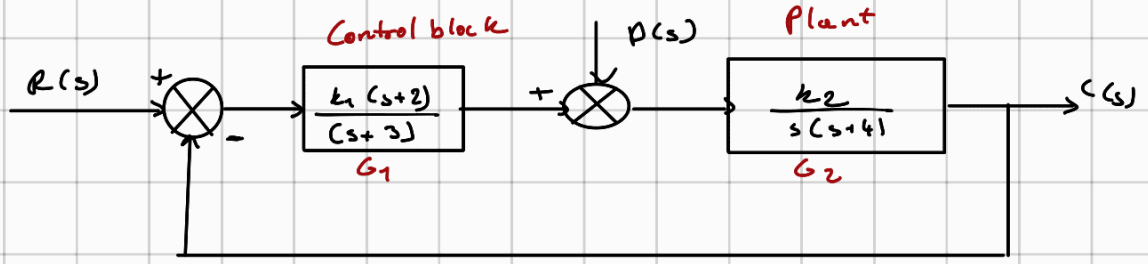
because it is negative we can decrease error with it

Example

Design k_1 and k_2 so that

1 - e_{ss} due to unit step disturbance : $-0,00001$

2 - e_{ss} " " ramp input : $0,002$



$$\frac{k_1(s+3-1)}{s+3} = \frac{k_1(s+3)}{s+3} - \frac{k_1}{s+3} = k_1 - \frac{k_1}{s+3}$$

$$e_{ss}^{D} = -\lim_{s\to 0} \frac{s\, G_2(s)\, D(s)}{1 + G_1(s)\, G_2(s)} = -\lim_{s\to 0} \frac{G_2(s)}{1 + G_1(s)\, G_2(s)} = -\frac{3}{2k_1}$$

$$-0,000001 = -\frac{3}{2k_1} \Rightarrow k_1 = 150\,000$$

$$e_{ss}^{R} = \lim_{s\to 0} \frac{sR(s)}{1 + G_1(s)\, G_2(s)} = \lim_{s\to 0} \frac{1}{s(1 + G_1(s)\, G_2(s))}$$

$$= \frac{6}{k_1 k_2} = 0,002 \Rightarrow k_2 = 0,02$$

$$\text{Total } e_{ss} = \frac{6}{k_1 k_2} - \frac{3}{2k_1}$$

$$= 0,02 - 0,000001$$

this system order is 3 by finding charackteristic equation.

Sensitivity

The degree to which changes in the system parameters affect, system T.F and hence performance of the system

(error, stability, ---)

Example:

$$F = \frac{k}{k+a} = \frac{10}{10+a}$$

if $k = 10$

$a = 100 \rightarrow F = \frac{10}{110} = 0,09$

$a = +200\% \rightarrow a = 300 \rightarrow F = \frac{10}{310} = 0,032$

$\frac{\Delta F}{F} \times 100 = \frac{0,032 - 0,091}{0,091} \times 100 = -65\%$

$\frac{\Delta a}{a} \times 100 = \frac{300-100}{100} \times 100 = 200\%$

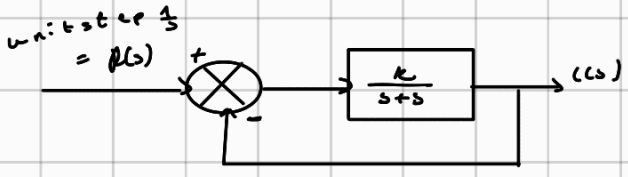
(Reduced sensitivity)
This means it's low sensitive to a

Average sensitivity: $\frac{65\%}{200\%} < 1$ (Reduced sensitivity)

$$S'_{F:a} = \lim_{\Delta p \to 0} \frac{\text{fractional change in function } (F)}{\text{// // // parameter } (p)}$$

$$= \lim_{\Delta p \to 0} \frac{\Delta F/F}{\Delta p/p} = \lim_{\Delta p \to 0} \frac{p}{F} \frac{\Delta F}{\Delta p}$$

$$S_{F:a} = \frac{p}{F} \frac{\partial F}{\partial p}$$



(check the sensitivity of ss. error w.r.t parameter 'k'

$$e(\infty) = e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1+G(s)} = \lim_{s \to 0} \frac{1}{1+G(s)} = \frac{1}{1+\lim_{s \to 0} \frac{k}{s+5}} = \frac{5}{5+k}$$

$$S'_{e:k} = \frac{k}{e_{ss}} \cdot \frac{\partial e_{ss}}{\partial k} = \left(\frac{k}{\frac{5}{5+k}}\right)\left(-\frac{5}{(5+k)^2}\right)$$

$$= \frac{k(5+k)}{5}\left(-\frac{5}{(5+k)^2}\right) = -\frac{k}{5+k}$$

for $k = 10 \Rightarrow S_{e:k} = -\frac{10}{5+10} = -\frac{10}{15} = -\frac{2}{3}$

If 'k' is changed to '12'

$\frac{12-10}{10} \leftarrow \frac{\Delta k}{k} \times 100 = 20\%$

δ → also

$$\underbrace{S_{e:k}}_{-\frac{2}{3}} \cdot \underbrace{\frac{\Delta k}{k}}_{\frac{1}{5}} = \frac{\delta e_{ss}}{e_{ss}} = -\frac{2}{3} \times \frac{1}{5} = -\frac{2}{15} = -13\%$$

$\delta_{ess} = -0,13 \times e_{ss}$

$= -0,13 \times \frac{1}{5} = -0,04$ the amount of change